Richie Mouhouadi | Paris, France

Builder – Data Scientist & AI – Small Industrial Field Experience

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French & English · Open to travelling

Profile

Former R&D reservoir engineer turned Machine Learning, Data Science Scientist & AI builder. I ship production systems on my own, work alongside operational teams to find the data no system holds and frame the work as business outcomes. I use AI to go fast, and I own the review: tests, CI, access control, secrets handling.

Experience

SUEZ Digital Solutions – Smart Metering Apprentice (Project & product support) *Paris, France · Feb 2025 – Mar 2027*

Prepaid smart water solution for emerging markets (Africa & Latin America):

- Support project manager on client contract delivery (Build & Run) for pilots in Senegal (5000 meters, pilots scaling toward national rollout) and Luanda, Angola (5000 meters).
- Extracted undocumented LoRaWan telemetry (RSSI, SNR, Gateways) from production MySQL JSON fields, validated field meanings with the product manager, and working on the network-health supervision Power BI Dashboard my manager requested.
- Working on a MLOps infrastructure for predictive maintenance of smart metering assets (Gateways, Transmitters, smart meters)
- Building a study on 10k+ households and ~3years of prepaid data from previous pilots, designed to support tenders: adoption, recharge behavior, consumption profiles, socio-economic impact on adoption behaviors.

Beicip-Franlab – Reservoir Engineer Intern *Ruel-Malmaison, France · Jan 2024 – June 2024*

Machine Learning applied to Reservoir Engineering:

- Trained a GAN on fewer than 130 images to reproduce the stratigraphic evolution sequence (formation thickness through time) obtained using Monte Carlo simulation, replacing a slow simulation run with fast generation.
- Forecast oil production with an LSTM. A graph neural network for Oilfield material-balance underperformed; I traced it to missing physical inputs (inter-well porosity, fluid flow mechanism,…) and documented what next model needed.

Chang'an University – R&D Reservoir Engineer *Xi'an, China · Sep 2020 – June 2023*

R&D in reservoir engineering, contributed to 3 scientific papers and 1 patent application:

- ORCID: https://orcid.org/0009-0007-2470-2110

Personal Projects – Solo, from zero, some shipped

BuilderLab – Team-delivery platform for builders/Learners (Live beta, 5 users – paused)

Live at: https://builderlab-tau.vercel.app/

Repo : https://github.com/rdmouhouadi/BuilderLab

Stack: Next.js, TypeScript, Supabase (PostgreSQL + Row Level Security), Vercel, Docker, GitHub Actions

- Code reviewed by DSTI's Scientific department before publication on the DSTI TechBlog (https://dsti.school/techblog/builderlab-shared-engineering-workflow), the review flagged missing tests and oversized components.
- Closed some issue raised from the code review: added GitHub Actions CI, lint errors 54 to 0, component decomposition (code refactoring), containerized build.
- Authorization enforced in the database (RLS policies, triggers for state transitions), service-role key kept server-side.

DocKA – Open-source document ingestion & search · v0.1.0
Repo : https://github.com/rdmouhouadi/Docka
Stack: Python, FastAPI, Elasticsearch, PostgreSQL, Airflow, Docker

- Ingests PDF, Word, HTML and text files into searchable knowledge base (BM25), with unit tests; roadmap to hybrid search and evaluated RAG.
- Data stays on the user's own infrastructure; nothing is indexed without explicit action.

StackPuls – Infrastructure alerts for Solo builders · In development (v0.6)
Landing page: https://stakpuls.com/
Repo: on demand (private github repo)
Stack: Typescript, Neon Postgres, Vercel, Groq, Cloudflare Workers

- LLM parsing of cloud alerts emails into one dashboard; reworking the setup flow before opening to users (Project on hold)

Education

MSc Data Science & AI – DSTI School of Engineer — *Paris, France, Feb 2025 – March 2027*
DevOps Bootcamp – TechWorld with Nana — *Online, Aug 2025 – Present*
MSc Geoenergies – Univerity of Pau and Adour Countries — *Pau, France, Sep 2023 – June 2024*

Skills

Data: Python, SQL, Power BI, Airflow

Cloud: AWS

Builder tools: TypeScripts, React/Next.js, FastAPI, Docker, GitHub Action, Vercel, Supabase